

Notes: Doshi and Kumar (RFS, 2025)

Capital Investment, Equity Returns, and Aggregate Dynamics in Oligopolistic Production Economies

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Calibration and aggregate data	4
2.2	How the quantitative sectors are constructed	5
2.3	Calibration targets and parameters	5
2.4	Empirical sample for cross-sectional return tests	6
2.5	Key empirical variables	6
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Consumer preferences and asset pricing	7
3.2	Production, capacity utilization, and productivity	7
3.3	Profits, dividends, and the multistage game	8
3.4	Inverse demand and the monopoly benchmark	8
3.5	Tacit collusion and the punishment equilibrium	9
3.6	Equilibrium characterization with a binding collusion constraint	9
3.7	Why real frictions matter more than in standard models	10
3.8	Quantitative implementation	10
3.9	Concentration, punishment effect, and market dilution effect	10
3.10	Role of operating leverage, profitability, depreciation sensitivity, and boom states	10
3.11	Empirical return regression	11
3.12	Empirical investment regression	11
4	Identification and instruments (what makes the estimates credible)	11
4.1	What standard empirical approaches miss	11

4.2	What identifies the quantitative model	12
4.3	What identifies the empirical hypotheses	12
4.4	Why off-equilibrium values are the distinctive identification idea	12
4.5	Empirical credibility and limitations	12
4.6	Relation to the literature	13
5	Tables and figures	13
5.1	Table 1: Calibration	13
5.2	Table 2: Financial and real moments	13
5.3	Table 3: Cyclical properties and real dynamics	14
5.4	Table 4: Effects of adjustment costs and productivity volatility	14
5.5	Figures 1 and 2: Impulse responses to productivity shocks	15
5.6	Figures 3, 4, and 5: Concentration and returns in industry equilibrium	16
5.7	Table 5: Concentration, operating leverage, and operating profitability	16
5.8	Table 6: Depreciation sensitivity and boom-state results	17
5.9	Table 7: Entry deterrence and capital investment	17
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion (what the paper establishes)	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** How does **tacit collusion** in oligopolistic industries with **capital investment** affect firm investment, capacity utilization, equity returns, and aggregate dynamics? Why is the empirical relation between **industry concentration** and **stock returns** sometimes positive and sometimes negative?
- **Setting.** The paper studies a **dynamic general equilibrium production economy** with multiple oligopolistic consumer-goods sectors, unlevered firms, capital accumulation, and real frictions such as **fixed operating costs**, **capital adjustment costs**, and extensions with **endogenous depreciation**. Firms sustain tacit collusion through a repeated-game punishment threat.
- **Core challenge.** Existing oligopoly asset-pricing models often abstract from capital investment, while production-based asset-pricing models usually assume competitive product markets. Without capital, one misses an important channel: **future stock values determine whether a firm wants to defect from collusion today**. This makes off-equilibrium firm values central for both real choices and returns.
- **Main theoretical contribution.** The paper shows that:
 1. tacit collusion links **real decisions** and **financial markets** because firms compare current gains from defection with future losses in equity value under punishment;
 2. capital and real frictions matter not only on the equilibrium path but also through their effect on the **off-equilibrium punishment path**;
 3. the concentration-return relation is **ambiguous in general**, and its sign depends on characteristics such as **operating leverage**, **operating profitability**, and **depreciation sensitivity**.
- **Why it matters.** The paper offers a unified way to think about three literatures at once:
 - dynamic oligopoly and tacit collusion,
 - production-based asset pricing,
 - empirical work on industry concentration and stock returns.

It also gives a reason why concentration might predict *lower* returns in some samples and *higher* returns in others.

- **Main quantitative design.** The model is calibrated in a **two-sector economy**:
 - a **concentrated sector** with tacit collusion,
 - a **low-concentration sector** with Bertrand–Nash price competition.

The two sectors are otherwise given **identical primitives**, so the comparison isolates the effect of tacit collusion.

- **Main quantitative results.**
 - Investment-rate volatility in the collusive sector is **15.31%** versus **6.25%** in the low-concentration sector.
 - Output-growth volatility is **6.12%** versus **2.50%**.

- The mean equity risk premium in the collusive sector is **0.71%**, versus essentially **0.00%** in the Bertrand benchmark.
- The volatility of the risk premium is **2.91%** in the collusive sector versus **0.20%** in the low-concentration sector.
- Lower adjustment costs have a **nonmonotonic** effect on the collusive-sector risk premium.
- Endogenous entry in the tacit-collusive sector can amplify both real and financial volatility.
- **Main empirical question.** Which firm and industry characteristics make the concentration-return slope negative versus positive in the data?
- **Main empirical results.**
 - Industry concentration by itself is **not significantly related** to returns in the full sample.
 - The interaction **concentration** $\times$ **operating leverage** is **negative and significant**: higher operating leverage makes the concentration-return relation more negative.
 - The interaction **concentration** $\times$ **operating profitability** is **positive and significant**: sufficiently profitable firms can have a positive concentration-return slope.
 - High-depreciation-sensitivity industries exhibit a **more negative** concentration-return relation.
 - In **boom quarters**, the negative concentration-return relation is stronger, and the interactions with operating leverage and profitability are also stronger.
 - In capital-intensive, high-entry-threat industries, capital investment is **positively related** to concentration, but this relation weakens after **2005**.
- **Economic takeaway.** Tacit collusion with capital creates a new asset-pricing channel: **returns depend on the credibility of punishment**. Because operating leverage, adjustment costs, and depreciation affect punishment values, they also shape whether concentration is associated with higher or lower expected returns.

2 Data and measurement (key objects)

2.1 Calibration and aggregate data

- **Industry data.** The calibration uses annual industry-level data on **capital, investment, output, and productivity** from the **NBER–CES manufacturing database**. Because productivity is available only through 2016, the calibration sample is **1958–2016**.
- **Capacity utilization and production.** The paper uses **Federal Reserve Board** series on manufacturing capacity utilization and consumer-goods production.
- **Aggregate quantities.** From **NIPA**, the paper uses:
 - real per capita personal consumption expenditures on durable and nondurable goods to discipline aggregate real consumption,
 - personal income from assets to discipline aggregate financial income.
- **Other macro data.**

- Population: **U.S. Census Bureau**
- Price deflator for personal consumption expenditures on goods: **BEA**
- Returns data: **Kenneth French data library**

2.2 How the quantitative sectors are constructed

- **Concentrated sector.** Consumer-goods manufacturing industries in the **top quartile** of concentration, where concentration is measured by the **sales share of the eight largest firms**.
- **Low-concentration sector.** Consumer-goods industries **below the median** concentration level.
- **Isolation strategy.** The two sectors are given:
 - identical production function,
 - identical cost parameters,
 - identical real frictions,
 - identical stochastic productivity process,
 - identical utility weights in the consumption basket.

Therefore, differences in outcomes come from **concentration and equilibrium conduct**, not from ex ante technological or preference differences.

- **Baseline number of firms in simulations.** The simulated economy uses **7 firms** in the concentrated sector and **54 firms** in the low-concentration sector.

2.3 Calibration targets and parameters

- **Externally chosen parameters.**
 - Annual discount factor: $\alpha = 0.99$
 - Risk aversion: $\gamma = 5$
 - Intertemporal elasticity of substitution: $\eta^{-1} = 1.9$
 - Intratemporal elasticity of substitution: $\sigma = 4.8$
 - Sector weights: $\phi = 0.5$ each
 - Depreciation rate: $\delta = 5\%$
 - Adjustment-cost parameter: $\varphi = 8$
 - Productivity persistence: $\rho = 0.94$
- **Internally calibrated parameters.** The paper chooses common values of:

$$h, \quad m, \quad \psi, \quad \lambda$$

to match four moments:

- volatility of aggregate real per capita consumption growth,

- volatility of the concentrated-sector investment rate,
- steady-state capacity utilization in the concentrated sector,
- steady-state ratio of profit rates between concentrated and low-concentration sectors.
- **Resulting baseline values.**
 - Output elasticity of capital: $\psi = 0.49$
 - Marginal cost: $h = 0.05$
 - Fixed cost: $m = 0.1$
 - Annual volatility of productivity shocks: $\lambda = 3\%$
 - Cross-sector productivity correlation: 0

2.4 Empirical sample for cross-sectional return tests

- **Returns.** Firm-level monthly returns come from **CRSP** and are aggregated to **quarterly returns**.
- **Firm characteristics.** Quarterly accounting variables come from **Compustat** and are **lagged one quarter** so that the empirical tests use information available to investors.
- **Concentration.** The paper uses the **Text-Based Network Industry Classification** Herfindahl–Hirschman Index of **Hoberg and Phillips (2016)**, denoted THHI.
- **Beta.** CAPM betas are taken from the **WRDS Betasuite**.
- **Sample restrictions.** Financial firms are excluded. The final empirical sample covers **1989–2019** and includes **8,892 unique nonfinancial firms**.

2.5 Key empirical variables

- **Operating leverage (OL).**

$$OL = \frac{SG\&A}{SG\&A + COGS}$$

using lagged quarterly data.

- **Operating profitability (OP).**

$$OP = \frac{\text{Revenue} - \text{COGS} - \text{SG\&A}}{\text{Market Equity}}$$

- **Capital investment intensity (CII).**

$$CII = \frac{\text{Capital investment}}{\text{Total assets}}$$

- **Boom periods.** Quarters with **market return above 10%**.
- **High depreciation sensitivity.** Industries classified using the slope from a regression of **depreciation rates on capacity utilization**, using BEA and FRB data.
- **High-entry threat industries.** Two-digit SIC industries identified using evidence on entry activity and capital intensity from the industrial organization literature.

3 Models and empirical specifications (all equations + interpretation)

3.1 Consumer preferences and asset pricing

The representative consumer-investor has recursive Epstein–Zin / Kreps–Porteus preferences:

$$U_t = \left[(1 - \alpha) C_t^{1-\eta} + \alpha E_t \left(U_{t+1}^{1-\gamma} \right)^{\frac{1-\eta}{1-\gamma}} \right]^{\frac{1}{1-\eta}},$$

where C_t is a CES aggregate of sectoral consumption.

The period budget constraint is

$$p_t \cdot c_t \leq q_t \cdot (d_t + s_t) - q_{t+1} \cdot s_t \equiv W_t,$$

so W_t is net financial income.

The induced consumption demand is

$$c_{jt}(p_t, W_t) = \frac{W_t}{P_t} \left(\frac{P_t \phi_j}{p_{jt}} \right)^\sigma,$$

with aggregate price index

$$P_t = \left[\sum_{j=1}^J \phi_j^\sigma p_{jt}^{1-\sigma} \right]^{1/(1-\sigma)}.$$

The stochastic discount factor is

$$\mathcal{M}_{t,t+1} = \alpha \left(\frac{C_{t+1}}{C_t} \right)^{-\eta} \left[\frac{U_{t+1}}{E_t(U_{t+1}^{1-\gamma})^{1/(1-\gamma)}} \right]^{\eta-\gamma},$$

and the asset-pricing condition is

$$\frac{s_t}{P_t} = E_t \left[\mathcal{M}_{t,t+1} \frac{d_{t+1} + s_{t+1}}{P_{t+1}} \right].$$

Interpretation. Households price equities in a standard production-based asset-pricing environment, but here the dividends and stock prices they value are shaped by oligopolistic tacit collusion.

3.2 Production, capacity utilization, and productivity

The representative firm in a sector produces according to

$$Y_t = A_t (u_t K_t)^\psi,$$

where:

- A_t is sectoral productivity,
- K_t is installed capital,

- $u_t \in [0, 1]$ is capacity utilization,
- $\psi \in (0, 1)$ is the output elasticity of capital.

Productivity follows

$$a_t = \rho a_{t-1} + \varepsilon_t, \quad a_t = \log A_t.$$

Capital evolves as

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Investment is subject to quadratic adjustment costs:

$$\Phi(I_t, K_t) = I_t + \frac{\varphi}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 K_t.$$

Interpretation. Capital matters twice. It is the productive state variable, and it is also strategic: future installed capacity affects whether collusion can be sustained.

3.3 Profits, dividends, and the multistage game

Firms compete on product prices within sectors. Given price p_t , variable cost h , and fixed operating cost m , dividends are

$$D_t = \Pi_t - \Phi(I_t, K_t) - mK_t.$$

The firm maximizes the present value of dividends:

$$V_t(\Omega_t) = \max_{p_t, I_t} \{D_t + E_t[V_{t+1}(\Omega_{t+1})]\},$$

where Ω_t collects the current aggregate and sectoral state.

Interpretation. The firm chooses price and investment jointly. This is crucial: product-market conduct today changes future capital, and future capital changes future punishment after a deviation.

3.4 Inverse demand and the monopoly benchmark

The consumption system implies sectoral inverse demand

$$p_t(Y_t^{ind}) = \phi(Y_t^{ind})^{-1/\sigma} (W_t P_t^{\sigma-1})^{1/\sigma},$$

where Y_t^{ind} is industry output.

The one-shot monopoly price is

$$\hat{p} = \left(\frac{\sigma}{\sigma - 1} \right) h.$$

Interpretation. Higher industry output lowers price. Therefore, changing capacity utilization is equivalent to changing effective quantity. Tacit collusion can then be implemented by deliberately operating below full capacity.

3.5 Tacit collusion and the punishment equilibrium

The paper studies the symmetric subgame-perfect equilibrium that maximizes firm value subject to no profitable deviation. The punishment path after defection is the **Bertrand–Nash** equilibrium, denoted by tildes.

Under the punishment path, firms use full capacity and invest according to a standard Euler condition. The key idea is that a firm comparing collusion and defection weighs:

- **current gain from defecting**, because it can undercut rivals and temporarily use full capacity,
- **future loss from punishment**, because the industry then reverts to low-value Bertrand competition.

The tacit-collusion incentive compatibility constraint is

$$\bar{D}_t - D_t \leq E_t \left[V_{t+1}(\Omega_{t+1}) - \tilde{V}_{t+1}(\Omega_{t+1}) \right].$$

Using the model's structure, this becomes

$$(p_t - h)A_t K_t^\psi (1 - u_t^\psi) \leq E_t \left[V_{t+1} - \tilde{V}_{t+1} \right].$$

Interpretation. Excess capacity is what makes collusion enforceable. If firms were already at full capacity, deviating would not increase current profits much less? Actually the paper's point is the reverse: collusion requires *not* using full capacity, so that deviation to full capacity is tempting but can still be deterred by the future loss in value.

3.6 Equilibrium characterization with a binding collusion constraint

Let ς_t be the multiplier on the collusion constraint. The first-order condition for capacity utilization is

$$\frac{\partial D_t}{\partial u_t} = \frac{\varsigma_t}{1 + \varsigma_t} \frac{\partial \bar{D}_t}{\partial u_t},$$

and the Euler equation for investment can be written as

$$\left[1 + \varphi \left(\frac{I_t}{K_t} - \delta \right) \right] (1 - \chi_t) = E_t \left[\frac{\partial V_{t+1}}{\partial I_t} \right] - \chi_t E_t \left[\frac{\partial \tilde{V}_{t+1}}{\partial I_t} \right].$$

Interpretation.

- Without a binding collusion constraint, investment obeys the standard marginal- q logic.
- With tacit collusion, investment also matters because it changes next period's **punishment value** $\tilde{V}_{t+1}$.
- Hence, off-equilibrium stock prices directly influence equilibrium real decisions.

This is the central innovation of the paper.

3.7 Why real frictions matter more than in standard models

The paper emphasizes that fixed costs and adjustment costs affect returns through two channels:

1. the usual production-side channel familiar from production-based asset pricing;
2. a new **tacit-collusion channel**, because those frictions alter the severity of the punishment equilibrium.

For example, high adjustment costs make it harder for firms to respond after collusion breaks down, which changes punishment values and therefore changes deviation incentives today.

Interpretation. Real frictions now affect risk not only because they make dividend growth more rigid, but also because they change how harsh a future price war would be.

3.8 Quantitative implementation

The model is solved using perturbation methods and quadratic approximations around the deterministic steady state. The collusive sector is compared with a low-concentration Bertrand benchmark, keeping all primitives the same.

Interpretation. This design isolates the *pure effect of tacit collusion* on investment volatility, risk premia, and aggregate comovement.

3.9 Concentration, punishment effect, and market dilution effect

The paper decomposes the effect of concentration into two opposing forces:

- **Market dilution effect (MDE).** More firms dilute per-firm profits for given capacity.
- **Punishment effect (PE).** Lower concentration can strengthen collusion by making the punishment after deviation more severe or more credible.

Schematically, the sign of the effect of the number of firms on equilibrium price satisfies

$$\Delta_N(p_t) \propto \Delta_N E_t[V_{t+1} - \tilde{V}_{t+1}] - (p_t - h)\Delta_N(\bar{Y}_t - Y_t).$$

Interpretation. The concentration-return slope is ambiguous because the **punishment effect** and the **market dilution effect** pull in opposite directions.

3.10 Role of operating leverage, profitability, depreciation sensitivity, and boom states

The paper derives the following comparative statics:

- **High operating leverage.** High fixed or adjustment costs make firm values low in bad states, weakening the punishment effect. Then the market dilution effect dominates, so the concentration-return relation becomes **more negative**.
- **High operating profitability / low operating leverage.** When firms remain valuable in bad states, the punishment effect stays important. Then the concentration-return relation can turn **positive**.

- **High depreciation sensitivity of capacity utilization.** When using capacity is especially costly through higher depreciation, punishment is weaker, again making the concentration-return relation more **negative**.
- **Boom states.** If the covariance between the SDF and aggregate returns is stronger in high-return states, concentration effects and their interactions with firm characteristics should be **stronger in booms**.

3.11 Empirical return regression

The main Fama–MacBeth specification is

$$R_{i,t} = \beta_{0t} + \beta_{1t}\text{THHI}_{i,t-1} + \beta_{2t}X_{i,t-1} + \beta_{3t}(\text{THHI}_{i,t-1} \times X_{i,t-1}) + \Gamma(\text{controls}_{i,t-1}) + \varepsilon_{i,t},$$

where X is alternatively:

$$\text{OL}, \quad \text{OP}, \quad I_{\delta(u) > \text{median}}, \quad \text{etc.}$$

The controls include the Fama–French three-factor characteristics and leverage.

Interpretation. The paper does not claim that concentration has one universal unconditional effect. Instead, it tests the model’s prediction that the concentration-return slope **systematically changes sign across observable firm and industry characteristics**.

3.12 Empirical investment regression

To test entry-deterrence predictions, the paper estimates panel regressions of capital investment intensity:

$$\begin{aligned} \text{CII}_{i,t} = & \alpha + \beta_1\text{THHI}_{i,t-1} + \beta_2(\text{THHI}_{i,t-1} \times \text{HighEntry}_{i,t-1}) \\ & + \beta_3(\text{THHI}_{i,t-1} \times \text{HighEntry}_{i,t-1} \times I_{t \geq 2005}) + \gamma(\text{controls}_{i,t-1}) + \varepsilon_{i,t}. \end{aligned}$$

Interpretation. If concentration helps incumbents deter entry through capacity, then concentration should be positively related to capital investment in capital-intensive, high-entry-threat industries, though this link may weaken in the era of “superstar firms.”

4 Identification and instruments (what makes the estimates credible)

4.1 What standard empirical approaches miss

- A simple regression of returns on concentration is not informative because the model predicts the **sign is ambiguous**.
- Standard production-based asset pricing abstracts from product-market collusion.
- Standard dynamic oligopoly analyses often abstract from capital and real frictions or do not connect them to asset pricing.

Key point. The paper’s contribution is not “concentration always raises/lowers returns.” It is that the effect depends on **observable state and firm characteristics** in a disciplined way.

4.2 What identifies the quantitative model

- The paper **internally calibrates** h, m, ψ, λ to four steady-state and volatility targets.
- Other parameters are disciplined by outside evidence from macro, asset-pricing, and industrial-organization literatures.
- Because the collusive and low-concentration sectors are otherwise identical, differences in moments are attributed to **equilibrium conduct**.

Interpretation. This is a structural counterfactual identification strategy: the Bertrand sector is the benchmark that isolates the contribution of tacit collusion.

4.3 What identifies the empirical hypotheses

- **Lagged characteristics** are used in all regressions.
- The concentration measure is **text-based** and product-focused, matching the model’s emphasis on product substitutability.
- The empirical exercises test **interaction effects**, not just levels.
- Industry depreciation sensitivity comes from external industry-level data, not the same stock-return regressions.
- Boom versus no-boom splits use aggregate market conditions, which allows a separate test of time variation.

Interpretation. The empirical strategy is best seen as a set of **theory-driven comparative-static tests**, rather than a clean causal estimate of the effect of concentration itself.

4.4 Why off-equilibrium values are the distinctive identification idea

The sharpest conceptual identification idea in the paper is that tacit collusion is enforced by

$$\text{future loss from punishment} = E_t[V_{t+1} - \tilde{V}_{t+1}],$$

so one must model both:

- the **equilibrium path**,
- the **off-equilibrium Bertrand path**.

Interpretation. Once this is true, anything that changes the punishment path—fixed costs, adjustment costs, depreciation, capacity, or aggregate conditions—can alter expected returns even if it seems like a purely “real” parameter.

4.5 Empirical credibility and limitations

- The return tests are run on a broad sample of nonfinancial U.S. firms from 1989–2019.
- The baseline return regressions include standard asset-pricing controls.

- The paper’s strongest empirical claims concern the **sign and interaction patterns** predicted by theory, not the exact size of one deep structural parameter.
- A limitation is that financial leverage is abstracted from in the theory, so some measured operating-leverage effects in the data may partially reflect latent financing effects.

4.6 Relation to the literature

- **Dynamic oligopoly / tacit collusion:** extends Abreu–Rotemberg–Saloner type logic to a production economy with capital.
- **Production-based asset pricing:** introduces product-market strategic interaction into a framework usually built under competition.
- **Concentration and returns:** provides a reconciliation of mixed empirical evidence by showing the slope depends on operating leverage, profitability, and state dependence.

5 Tables and figures

5.1 Table 1: Calibration

Read:

- This is the quantitative backbone of the paper.
- The key values are:

$$\alpha = 0.99, \quad \gamma = 5, \quad \eta^{-1} = 1.9, \quad \sigma = 4.8, \quad \psi = 0.49, \quad h = 0.05, \quad m = 0.1,$$

$$\varphi = 8, \quad \delta = 5\%, \quad \lambda = 3\%, \quad \rho = 0.94.$$

- The interesting methodological point is that the two sectors are deliberately made **ex ante identical**, so the concentrated versus low-concentration comparison is very clean.

5.2 Table 2: Financial and real moments

Read:

- Collusion strongly amplifies the volatility of firm-level real activity.
- Investment-rate volatility: **15.31** in the concentrated sector versus **6.25** in the low-concentration sector.
- Output volatility: **6.12** versus **2.50**.
- Mean risk premium: **0.71%** versus essentially **0.00%**.
- Risk-premium volatility: **2.91%** versus **0.20%**.
- Aggregate consumption volatility is matched well (**2.49** in model versus **2.56** in data), and the mean real risk-free rate is also close (**1.02%** versus **0.91%**).

Table 1
Calibration

	Consumption and production
Annual discount factor (α)	0.99
Risk aversion (γ)	5
Intertemporal elasticity of substitution (η^{-1})	1.9
Intratemporal (product) elasticity of substitution (σ)	4.8
Sectoral utility weight (ϕ)	0.5
Output elasticity of capital (ψ)	0.49
Marginal cost (h)	0.05
Fixed cost (m)	0.1
Capital adjustment cost (ρ)	8
Annual depreciation rate (δ)	5%
Variance-covariance matrix of productivity shocks (annual)	
Volatility of productivity shocks ε_j (λ_{jj})	3%
Correlation of productivity shocks (λ_{ij})	0.0
Autocorrelation coefficients (annual)	
Coefficient for ε_j (ρ_j)	0.94

This table displays the calibration for simulations of the model with identical parameterization for the concentrated and low-concentration sectors.

Table 2
Financial and real moments

A. Sectoral real moments				
	SD (log changes)			
	Model		Data	
Investment rate: Concentrated sector (g_{Ir1})	15.31		15.91	
Investment rate: Low-concentration sector (g_{Ir2})	6.25		14.27	
Output: Concentrated sector (g_{y1})	6.12		21.86	
Output: Low-concentration sector (g_{y2})	2.50		12.13	
B. Sectoral financial moments				
	Mean		SD	
	Model	Data	Model	Data
Risk-premium: Concentrated sector (ERP_1)	0.71	6.94	2.91	19.19
Risk-premium: Low-concentration sector (ERP_2)	0.00	6.89	0.20	16.04
C. Aggregate real moments				
	SD (log changes)			
	Model		Data	
Aggregate real consumption (g_c)	2.49		2.56	
Aggregate financial income (g_w)	3.39		6.29	
Aggregate price index (g_π)	2.08		2.89	
D. Aggregate financial moments				
	Mean		SD	
	Model	Data	Model	Data
Real risk-free rate (r_f^r)	1.02	0.91	0.89	2.15

This table presents the moments of salient firm-level and aggregate real and financial variables from simulations of the model with the calibration specified in Table 1, and with 7 firms in the concentrated sector and 54 firms in the low-concentration sector. The displayed moments are based on 590,000 yearly simulations.

Takeaway. Tacit collusion materially strengthens both the real-financial link and priced risk exposure.

5.3 Table 3: Cyclical properties and real dynamics

Read:

- The collusive sector tends to produce cyclical comovement and autocorrelation patterns that are **closer to the data** than the Bertrand benchmark.
- The model also produces positive first-order autocorrelation in aggregate consumption growth, broadly consistent with the data.

Takeaway. Beyond average moments, tacit collusion improves the model's fit for dynamics.

5.4 Table 4: Effects of adjustment costs and productivity volatility

Read:

- Lowering the adjustment-cost parameter from $\varphi = 8$ to $\varphi = 5$ raises collusive-sector mean ERP from **0.71%** to **0.77%**.
- Lowering it further to $\varphi = 2$ lowers mean ERP to **0.67%**.
- This is the paper's quantitative evidence that the adjustment-cost / return relation is **nonmonotonic**.
- Lower productivity-shock volatility lowers the collusive-sector ERP and lowers aggregate real volatilities.

Takeaway. Real frictions matter for returns in a way that is more complicated than in competitive models because they change both equilibrium and punishment paths.

Table 3
Cyclical properties and real dynamics

A. Sectoral business cycle correlations				
	Concentrated sector		Low-concentration sector	
	Model	Data	Model	Data
Corr($I R, W$)	0.27	0.65	-0.17	0.94
Corr($I R, C$)	0.29	0.55	-0.02	0.88
Corr(Y, W)	0.85	0.98	0.99	0.95
Corr(Y, C)	0.92	0.96	0.99	0.96
B. Sectoral autocorrelation coefficients (log changes)				
	Concentrated sector		Low-concentration sector	
	Model	Data	Model	Data
Output (g_Y)				
ACF(1)	-0.50	-0.52	0.29	-0.03
ACF(2)	0.16	0.31	-0.16	-0.18
ACF(3)	-0.06	-0.19	-0.005	0.15
Investment rate (g_I)				
ACF(1)	0.06	-0.10	-0.67	0.03
ACF(2)	-0.03	-0.04	0.22	-0.09
ACF(3)	0.00	0.01	-0.07	-0.18
C. Autocorrelation coefficients of aggregate consumption growth (g_C)				
	Model	Data		
ACF(1)	0.29	0.25		
ACF(2)	-0.16	0.05		
ACF(3)	-0.005	-0.02		

Panel A and B analyze the cyclical properties and dynamics of real variables in the concentrated and low-concentration sectors. Panel C reports the dynamics of aggregate real consumption growth from simulations of the model with the calibration specified in Table 1 and sectoral concentrations specified in Table 2. The displayed quantities are based on 590,000 yearly simulations.

Table 4
Effects of varying adjustment costs and productivity shock process

A. Sectoral mean and standard deviation of equity risk premium			
	Lower adjustment costs (φ)		Lower shock volatility (λ)
	$\varphi = 5$	$\varphi = 2$	
Mean (%)			
Risk-premium: Concentrated sector (ERP_1)	0.77	0.67	0.58
Risk-premium: Low-concentration sector (ERP_2)	0.00	0.00	0.00
Standard deviation (%)			
Risk-premium: Concentrated sector (ERP_1)	2.45	1.60	2.49
Risk-premium: Low-concentration sector (ERP_2)	0.28	0.35	0.18
B. Aggregate real moments			
	Standard deviation - log changes (%)		
	$\varphi = 5$	$\varphi = 2$	λ
Aggregate consumption (g_C)	2.38	2.33	2.24
Aggregate income (g_W)	3.09	2.89	3.05
Aggregate price index (g_π)	1.96	1.89	1.88

This table analyzes the effects of varying adjustment costs and the volatility of the productivity shock on sectoral financial and aggregate real moments in the baseline model analyzed in Table 2. The displayed moments are based on 590,000 yearly simulations.

5.5 Figures 1 and 2: Impulse responses to productivity shocks

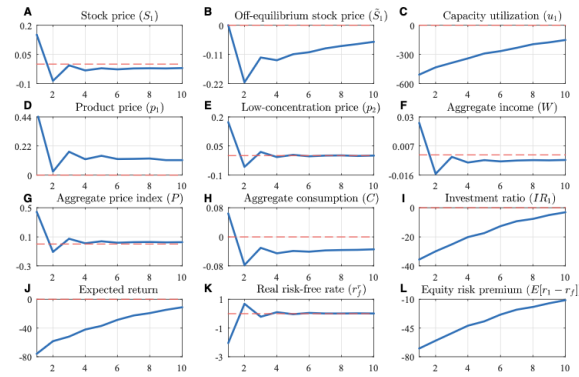


Figure 1
Responses to concentrated sector productivity shocks
This figure displays 10-year impulse response functions of salient firm-level and aggregate variables to a (one-standard-deviation) positive productivity shock in the concentrated sector. Unless specified, the firm-level variables refer to the concentrated sector. We scale all impulse responses by a factor of 10,000.

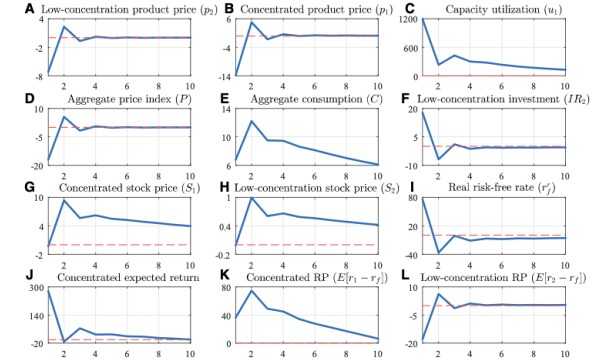


Figure 2
Responses to low-concentration sector productivity shocks
This figure displays 10-year impulse response functions of salient firm-level and aggregate variables to a (one-standard-deviation) positive productivity shock in the low-concentration sector. The sectoral location (concentrated or low-concentration sector) of firm-level variables is typically identified, except for capacity utilization, which varies only in the concentrated sector. RP refers to the risk premium. We scale all impulse responses by a factor of 10,000.

Read:

- Figure 1 studies a positive productivity shock in the **concentrated sector**.
- The shock raises the collusive-sector equilibrium stock price S_{1t} but lowers the punishment stock price $\tilde{S}_{1t}$.
- Because punishment becomes more severe in value terms, the collusion constraint is relaxed and capacity utilization falls toward the monopoly level.
- The strategic effect on investment can dominate the efficiency effect, so investment in the collusive sector can fall even after a positive productivity shock.
- Figure 2 studies a positive productivity shock in the **low-concentration sector**. It lowers the low-concentration sector price directly, tightens the collusion constraint in the concentrated

sector, raises collusive-sector utilization, and affects aggregate prices, income, and risk-free rates through general equilibrium.

Takeaway. These figures are the clearest visualization of the paper’s mechanism: **productivity shocks move stock prices, stock prices move collusion incentives, and collusion incentives move real decisions.**

5.6 Figures 3, 4, and 5: Concentration and returns in industry equilibrium

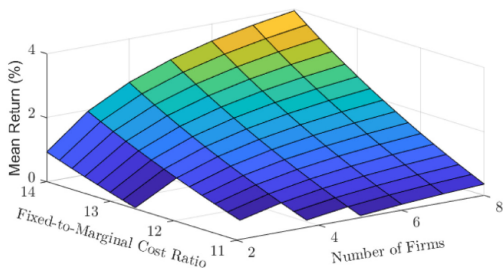


Figure 3
Industry concentration and equity returns with high operating leverage
This figure displays, through a three-dimensional surface plot, the relation of mean equity returns and industry concentration (inversely related to the number of industry firms) for high ratios of fixed-to-marginal costs, or high (operating leverage/operating profits) ratios, in the industry equilibrium analyzed in Section 4.1. The mean returns are computed from 1,000 simulations of 59 yearly observations.

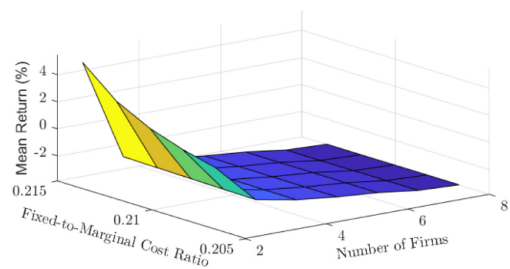


Figure 4
Industry concentration and equity returns with low operating leverage
This figure displays, through a three-dimensional surface plot, the relation of mean equity returns and industry concentration (inversely related to the number of industry firms) for low ratios of fixed-to-marginal costs, or high (operating profits/operating leverage) ratios, in the industry equilibrium analyzed in Section 4.1. The mean returns are computed from 1,000 simulations of 59 yearly observations.

Read:

- **Figure 3:** with **high operating leverage**, mean returns are monotonically **negatively** related to concentration.
- **Figure 4:** with **low operating leverage / high profitability**, the relation turns **positive**.
- **Figure 5:** with **high adjustment costs**, the concentration-return relation is again negative.

Takeaway. These figures are the paper’s theoretical bridge to the empirical literature: the sign of the concentration-return slope is not fixed.

5.7 Table 5: Concentration, operating leverage, and operating profitability

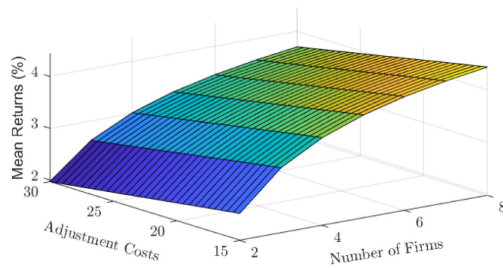


Figure 5
Industry concentration and equity returns with high adjustment costs
This figure displays, through a three-dimensional surface plot, the relation of mean equity returns and industry concentration (inversely related to the number of industry firms) for high levels of adjustment costs in the industry equilibrium analyzed in Section 4.1. The mean returns are computed from 1,000 simulations of 59 yearly observations.

Table 5 Industry concentration, stock returns, and firm characteristics								
Beta	ln(ME)	BE/ME	Leverage	THHI	OL	THHI x OL	OP	THHI x OP
0.61 (0.50)	−0.46*** (0.16)	1.64*** (0.47)	0.14 (1.42)					
0.55 (0.48)	−0.49*** (0.17)	1.62*** (0.47)	0.12 (1.42)	−0.87 (0.57)				
0.44 (0.45)	−0.41*** (0.15)	1.70*** (0.58)	1.17 (1.23)	0.56 (0.60)	3.45* (1.49)	−4.64* (2.23)		
0.75 (0.47)	−0.66*** (0.16)	1.42** (0.58)	−5.65*** (1.28)	−2.34*** (0.65)			62.64*** (5.95)	36.74*** (11.51)

This table analyzes the effects of operating leverage (OL) and operating profitability (OP) on the relation of industry concentration and stock returns at the firm level using the predictive cross-sectional [Fama and MacBeth \(1973\)](#) regressions (see Equation (21)). The Fama-MacBeth cross-sectional regressions of returns on firm characteristics are run each quarter. The table reports the time-series mean of the estimated coefficients. The number in the parentheses reports the corresponding Newey-West adjusted standard error. The sample period is from 1989 to 2019. Beta indicates CAPM beta; ME indicates the market value of equity; BE indicates the book value of equity; leverage indicates the market leverage; and THHI indicates the Text-based network industry classifications Herfindahl-Hirschman index of [Hoberg and Phillips \(2016\)](#). The appendix defines each variable and provides details of sample construction.

Read:

- Concentration alone (THHI) is insignificant in the full sample.
- The interaction $\text{THHI} \times \text{OL}$ is **negative and significant**.
- The interaction $\text{THHI} \times \text{OP}$ is **positive and significant**.
- The paper reports that the concentration-return relation is negative for firms with OL above the sample mean, and positive for firms with $\text{OP} > 6.4\%$.

Takeaway. This is the key reduced-form evidence that resolves the mixed concentration-return literature.

5.8 Table 6: Depreciation sensitivity and boom-state results

Read:

- High-depreciation-sensitivity industries have a **more negative** concentration-return relation.
- In boom quarters, concentration is significantly negatively related to returns.
- The interaction effects of concentration with operating leverage and profitability are stronger in boom quarters than in normal periods.

Takeaway. The model's state-dependent predictions survive in the data.

Table 6
Industry characteristics and time variation in effects of concentration on stock returns

Beta	ln(ME)	BE/ME	Leverage	THHI	THHI x OL	OL	THHI x OL	OP	THHI x OP
$\mathbb{I}_{\delta_{nd}(u) > \text{med}(\delta_{nd}(u))}$									
A. High depreciation slope									
0.43 (0.51)	-0.44** (0.18)	2.83*** (0.78)	0.25 (1.72)	-1.45* (0.81)	-1.07** (0.52)				
B. Boom quarters									
7.18*** (1.47)	-1.44*** (0.54)	1.62 (1.06)	0.75 (4.81)	-4.70** (1.96)					
6.35*** (1.30)	-1.05** (0.45)	1.56 (1.03)	6.97* (3.89)	3.11 (2.51)		16.50** (6.62)	-24.33** (10.95)		
6.95*** (1.42)	-1.41*** (0.50)	1.24 (1.14)	-2.18 (4.12)	-8.21*** (2.22)				31.31** (15.60)	89.55*** (30.37)
C. No-boom quarters									
-0.64 (0.40)	-0.31** (0.15)	1.62*** (0.51)	-0.00 (1.23)	-0.18 (0.45)					
-0.63 (0.40)	-0.30** (0.14)	1.72*** (0.65)	0.12 (1.15)	0.10 (0.58)		1.09 (1.04)	-1.08 (1.51)		
-0.37 (0.41)	-0.52*** (0.14)	1.45** (0.64)	-6.28*** (1.19)	-1.28** (0.53)				68.30*** (6.18)	27.19** (12.26)

Panel A of this table analyzes the relationship between industry concentration and stock returns at the firm level for industries with high depreciation costs of capacity utilization (identified as a dummy variable). $\delta_{nd}(u)$ indicates the slope from the regression of industry depreciation rates on industry capacity utilization. $\mathbb{I}_{\delta_{nd}(u) > \text{med}(\delta_{nd}(u))}$ is set to one if the industry slope is above the median. Panels B and C analyze time variation in the relation of industry concentration and stock returns at the firm level. Panel B presents the time-series mean of the estimated coefficients for quarters in which the market return exceeds 10% while Panel C presents the time-series mean of the estimated coefficients for the remaining quarters. Predictive cross-sectional Fama and MacBeth (1973) regressions (see Equation (21)) are utilized for these analyses. The Fama-MacBeth cross-sectional regressions of returns on firm characteristics are run each quarter. The table reports the time-series mean of the estimated coefficients. The number in the parentheses reports the corresponding Newey-West adjusted standard error. The sample period is from 1989 to 2019. Beta indicates CAPM beta; ME indicates the market value of equity; BE indicates the book value of equity; leverage indicates the market leverage; and THHI indicates the Text-based network industry classifications Herfindahl-Hirschman index of Hoberg and Phillips (2016). The appendix defines each variable and provides details of sample construction.

Table 7
Entry deterrence and capital investment

q	ln(ME)	Leverage	THHI	High entry	THHI x High entry	(THHI x High entry) $\mathbb{I}_{\geq 2005}$
-0.0002 (0.0002)	0.03*** (0.01)	-1.06*** (0.06)	-1.09*** (0.06)	-1.18*** (0.04)	1.21*** (0.07)	
-0.0002 (0.0002)	0.03*** (0.01)	-1.06*** (0.06)	-1.09*** (0.06)	-1.18*** (0.04)	1.26*** (0.07)	-0.12* (0.07)

This table analyzes the effects of strategic entry deterrence on capital investment intensity (CII) and the time variation in these effects. We run the following panel regressions:

$$CII_{i,t} = \alpha + \beta_1 THHI_{i,t-1} + \beta_2 THHI_{i,t-1} * Highentry_{i,t-1} + \gamma Controls_{i,t-1} + \epsilon_{i,t},$$

$$CII_{i,t} = \alpha + \beta_1 THHI_{i,t-1} + \beta_2 THHI_{i,t-1} * Highentry_{i,t-1}$$

$$+ \beta_3 THHI_{i,t-1} * Highentry_{i,t-1} * \mathbb{I}_{\geq 2005} + \gamma Controls_{i,t-1} + \epsilon_{i,t}.$$

For each estimated coefficient, the number in the parentheses reports the corresponding robust standard errors clustered at the firm level. $Controls_{i,t-1}$ represents the vector of controls that include Tobin's q indicated by q, the market value of equity indicated by ME, the market leverage, and the year dummies. THHI indicates the Text-based network industry classifications Herfindahl-Hirschman index of Hoberg and Phillips (2016). High entry identifies capital-intensive high entry threat industries, and $\mathbb{I}_{\geq 2005}$ indicates a dummy equal to one from 2005 onwards. The appendix defines each variable and provides details of sample construction. For ease of tabulation, the dependent variable is scaled by 100. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.9 Table 7: Entry deterrence and capital investment

Read:

- In capital-intensive, high-entry-threat industries, concentration is **positively related** to capital investment.
- The positive concentration-investment relation becomes **weaker after 2005**.

Takeaway. This supports the idea that concentration can reflect strategic capacity building for entry deterrence, but that this mechanism is weaker in the more recent “superstar firm” era.

6 Short summary

- **Method.** Build a dynamic production economy with oligopolistic sectors, capital investment, and tacit collusion enforced by a future punishment path.
- **Key mechanism.** A firm’s no-defection condition depends on **equilibrium** and **off-equilibrium** stock values, so financial markets directly affect real decisions.
- **Quantitative result.** Relative to an otherwise identical Bertrand sector, tacit collusion amplifies investment volatility, output volatility, and the mean and volatility of the equity risk premium.
- **Empirical result.** Concentration alone has little predictive power for returns, but its interaction with operating leverage, operating profitability, depreciation sensitivity, and boom states matches the model’s predictions.
- **Main lesson.** The concentration-return relation is not one-dimensional. It depends on how real frictions shape the severity of punishment after collusive breakdown.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Tacit collusion in a production economy with capital investment generates a strong link between **real decisions**, **financial markets**, and **aggregate dynamics**.
- Off-equilibrium stock prices are not a technical side issue; they are central because they discipline defection from collusion.
- Once capital and real frictions are present, the effect of concentration on expected returns becomes ambiguous in theory but predictable conditional on firm characteristics.

7.2 Economic intuition

- A more concentrated industry can have higher markups, but concentration also changes how credible and painful future punishment is after deviation.
- Firms with high operating leverage or high adjustment/depreciation sensitivity have weak punishment values in bad states, so concentration tends to predict lower returns there.
- Firms with low operating leverage or high operating profitability keep more value in punishment states, so concentration can instead predict higher returns.

7.3 Takeaway

This paper is important because it shows that **industrial organization and asset pricing cannot be cleanly separated** when firms invest and collude dynamically. The right way to think about

returns in concentrated industries is not simply “more market power means higher profits.” The relevant question is: **how does concentration interact with capital, real frictions, and punishment values?** Once that is recognized, both the quantitative and empirical results become much easier to understand.